

Online supplement to:
*Completion-aware search with partial witness repair for dense
fixed-shape facility layout*

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This supplement contains four secondary experiments referenced from the main text: zero shot transfer (S1), quality and diversity fronts (S2), amortized cost (S3) and action granularity (S4), followed by the full improvement formulation ablation (S5), a qualitative certificate gallery (S6), the complete MCNC/GSRC conversion (S7) and the large instance feasibility record (S8). Each experiment follows the protocol in the main text. These studies provide complementary diagnostics, with a relevant control for each experiment. Section S9 gives the full budget sensitivity and mechanism record for the comparison in which all methods start from the same layout. Section S10 collects additional formulation and certificate diagnostics; complete run-level outputs remain in the archive. Sections S11 and S12 document secondary methods, dataset coverage and policy training. Section S13 reports the 60-second comparison of M0, M1, and ALNS, including configuration selection, coverage, quality curves, target attainment, and timing checks.

S1 Zero shot transfer

A trained policy can have an advantage through amortization. A solver restarts for every brief, while a policy evaluates a new brief in one forward pass. We trained a policy only on procedurally generated briefs. The type inventory, dimensions, affinity tables, plate aspect ratio, fill ratio, wall preferences and objective constants were randomized during 1.2M training steps. We then evaluated the policy on briefs that were not optimized during training.

Table S1: Zero shot transfer at $n = 32$. The greedy constructor provides the direct comparison because, like the policy, it does not search over the plate. It also requires no training. Its cost is 32 and 528 marginal objective sweeps per layout; the batched figure on the heterogeneous generated set is higher, so we report the cost for one instance.

test set	shelf	SA 45k lat.	greedy constructive		RL zero shot ($\times 16 + \text{pol.}$)
			area ord.	room+pos.	
generated (in distribution)	1 912.6	2 613.9	2 488.8	2 597.5	2 591.0
OFFICE (unseen)	2 942.5	3 412.1	3 359.5	3 380.2	3 426.3
CLINIC (unseen)	3 503.7	4 011.9	3 955.6	4 116.5	3 950.2
evaluations/layout	1	58 963 to 65 875	32	528	2 049 to 5 633

On OFFICE the zero shot policy reaches 3 414.5 with a single greedy decode (32 evaluations, 12.7 ms), matching annealing’s 3 412.1 at 60 371 evaluations, and 3 426.3 with 16 samples and a polish. At the same 32 evaluations the constructor without training reaches 3 359.5.

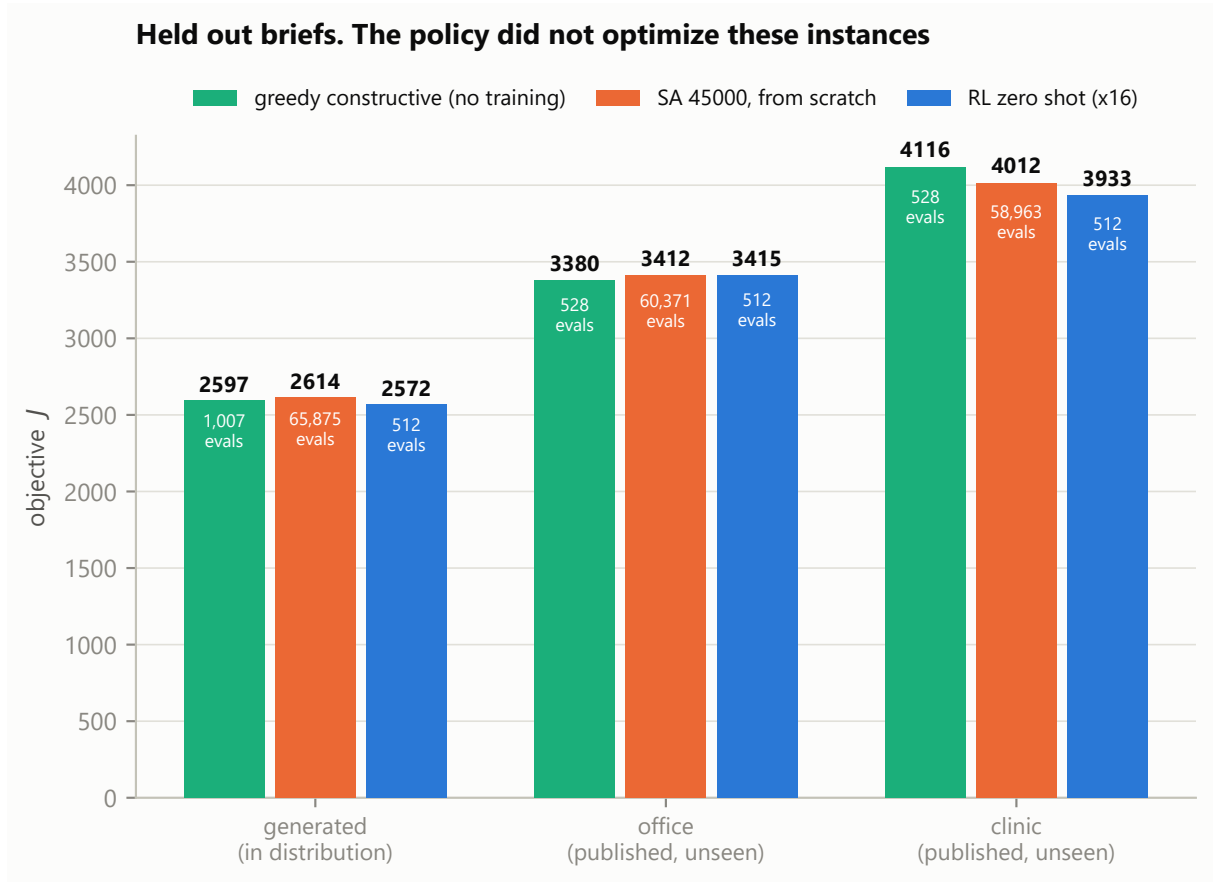


Figure S1: Held out briefs, annotated with objective evaluations per layout. The two reference scenarios were authored by hand and never optimized in training. The RL bars show 16 samples without the final polish (512 evaluations), a cheaper operating point than the $\times 16$ +polish column of Table S1, which is why the two differ slightly.

CLINIC gives a different result. The greedy constructor reaches 4 116.5, which is higher than both full budget annealing and the zero shot policy. At a matched 32 evaluations, it reaches 3 955.6 against the policy’s 3 848.5. Across the three test sets the zero shot policy is better on one, similar on one and worse on one. These results show that one set of weights can evaluate unseen briefs with a quality similar to large budget annealing while using three to four orders of magnitude less search, with the relative ranking varying across briefs. Transfer is evaluated at a fixed facility count (main text, Limitations).

S2 Quality and diversity

A common argument for learning in design is that a policy provides a *distribution* over layouts. To compare quality and diversity across all three methods, we vary the sampling temperature for the policy, the budget for annealing, and the fraction of randomized commitments for the constructor without training.

Table S2: Quality against diversity on OFFICE, 64 samples from one start. Diversity is the mean pairwise fraction of cells whose room type differs, estimated from 400 of the 2016 pairs, so it carries about ± 0.01 of sampling noise; every row uses the same estimator.

method	J	diversity	ms/sample
policy, order chosen by policy, $T=1.0$	3 529.3	0.006	22
policy, random commit order, $T=1.0$	3 508.0	0.304	19
$T=2.5$	3 418.2	0.405	18
$T=5.0$	3 048.0	0.623	13
SA 8 192, lattice	3 321.1	0.581	478
SA 45 000, lattice	3 414.5	0.639	1 070
greedy constructive, area order	3 359.5	0.000	7
20% of commit slots reshuffled	3 374.5	0.350	114
50%	3 408.3	0.482	251
70%	3 415.2	0.531	294
100%	3 405.2	0.620	347

We make three observations. First, a policy whose entropy bonus is annealed to zero produces almost no diversity. At $T=1$, its diversity is 0.006. Second, randomizing the commit order increases diversity by a factor of fifty with a 0.6% decrease in objective. Third, applying the same procedure to the greedy rule produces its own front. Randomizing a controlled fraction of the commit slots changes diversity continuously from 0.000 to 0.620 and reaches 3 415.2 at diversity 0.531 for 32 objective evaluations, which matches the best annealing point anywhere on the front (3 414.5 at 0.639, 61 267 evaluations).

The two fronts cross once. The policy has a higher objective up to diversity ≈ 0.42 , while the constructor without training has a higher objective above ≈ 0.45 . Partial randomization of the commit order resolves the interval between 0.000 and 0.603 left between the deterministic and uniformly randomized settings. This comparison covers a single scenario; evaluation across the generated suite remains future work.

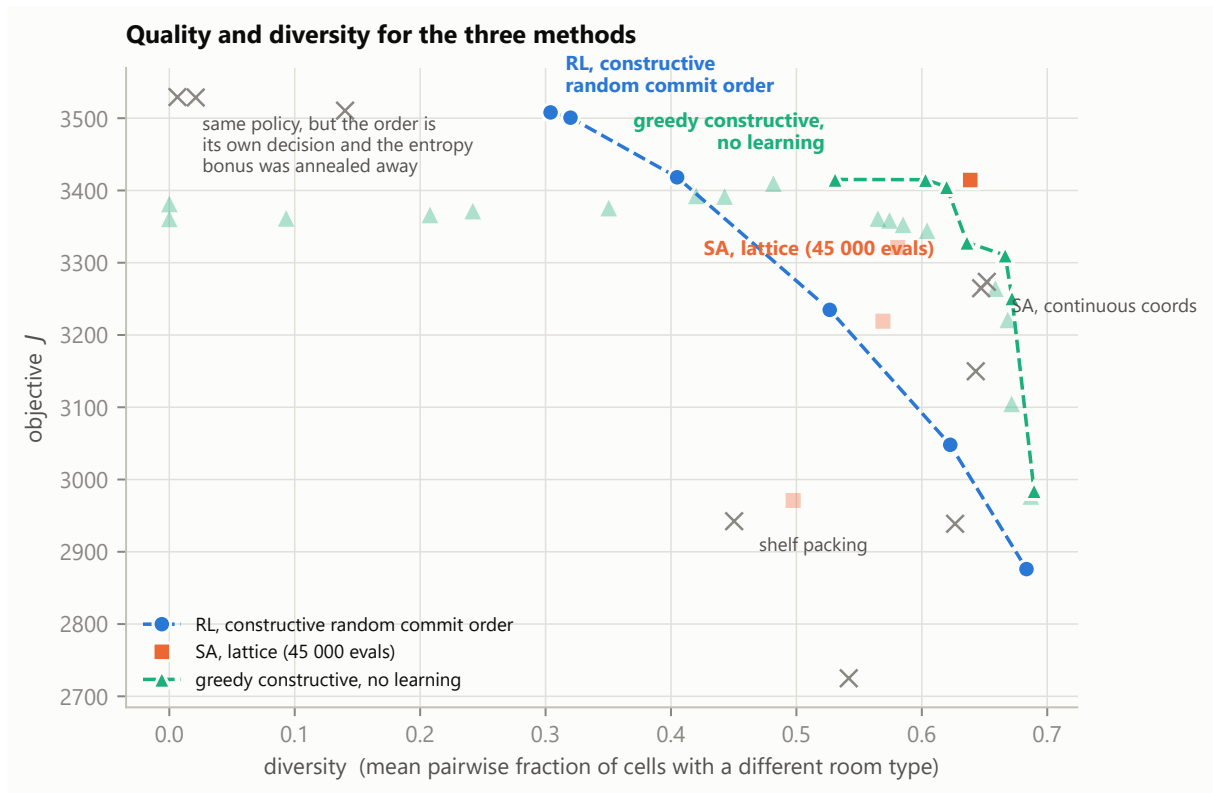


Figure S2: Quality against diversity with all three sides swept. Faint markers are points that a method's own front dominates.

S3 Amortized cost

Training has a fixed initial cost, while search has a cost for every layout. Table S3 reports the annealing budget sweep on OFFICE, each budget’s wall clock time per layout, and the first point on the policy’s training curve (488 recorded checkpoints, 2 049s total) at which the policy’s greedy decode exceeds that budget’s mean objective. The policy finishes at 3 529.3 and decodes in 17.8ms per layout.

Table S3: Annealing budget against the training time needed to match it. The 45k lattice row is the measurement of the scenario comparison (the budget sweep’s own 45k run gives 3 400.8 at 1.13s); the larger budgets come from the dedicated sweep. The reference scenario implementation is not batched, so its time per layout is omitted.

annealer	mean J	s/layout	training s to exceed
non-batched reference, 45k	3 269.9	not reported	109
lattice, 45k	3 414.8	1.00	157
lattice, 150k	3 459.0	3.40	215
lattice, 400k	3 489.1	8.83	256
lattice, 10^6	3 501.3	20.08	268

Charging the full 2 049s up front and 17.8ms per layout thereafter, the policy has a lower total wall clock cost than annealing with 10^6 evaluations after $2049/(20.08-0.018) \approx 102$ layouts. Compared with lattice annealing using 45 000 evaluations, this occurs after $2049/(1.00-0.018) \approx 2 093$ layouts.

This comparison has two limitations. First, the workload contains repeated requests for the *same* brief. An annealer could store its best layout and reuse it for later requests. A stream of unseen briefs is evaluated separately in S1. Second, this comparison is against annealing and not against the constructor without training. At 88ms per layout, the initial training cost is recovered only after roughly 29 000 layouts relative to the constructor. For smaller workloads, this constructor has the lower total cost.

S4 Action granularity

We compare a discrete action head for small moves with a continuous displacement head when feasibility is handled by a penalty. We report the comparison inside the improvement MDP on the single $n = 32$ OFFICE scenario (best observed reward, jump and swap moves, Metropolis acceptance). One checkpoint per action head is evaluated from the same 64 independent random starts, shown in Table S4. No tradeoff appears in this evaluation. With sampled decoding at horizon 4 096 the lattice head reaches 3 069.6 at 50% feasibility while the continuous head reaches 2 737.0 at 0%, and deterministic decoding of either head falls into a limit cycle and loses more than a thousand objective units. The feasible-volume record in Table S12 explains the direction geometrically, and under the constructive formulation the question does not arise.

S5 Improvement formulation ablation, in full

The main text summarizes the improvement MDP variants on OFFICE; Table S5 gives every variant. Each is trained for a matched budget of 688k environment steps and evaluated on 64

Table S4: Discrete against continuous action head inside the improvement MDP with best observed reward, jump and swap moves, and Metropolis acceptance on OFFICE, $n = 32$: mean and standard deviation across the same 64 independent evaluation starts.

head	decode	horizon	J	feasible
discrete (lattice)	sampled	512	$2\,475.3 \pm 93.8$	0.00
discrete (lattice)	sampled	2\,048	$2\,853.0 \pm 101.4$	0.06
discrete (lattice)	sampled	4\,096	$3\,069.6 \pm 126.3$	0.50
discrete (lattice)	argmax	4\,096	$1\,372.6 \pm 228.4$	0.00
continuous	sampled	512	$2\,354.7 \pm 76.2$	0.00
continuous	sampled	2\,048	$2\,611.4 \pm 85.5$	0.00
continuous	sampled	4\,096	$2\,737.0 \pm 60.8$	0.00
continuous	mean	4\,096	$1\,649.5 \pm 203.9$	0.00

held out starts using stochastic decoding at 4\,096 objective evaluations.

Table S5: Improvement formulation ablation on OFFICE.

ID	configuration	J	feasible
A0	reference: difference reward, translate, $\gamma=0.995$	2\,658.5	0 %
A0'	the same with $\gamma=0$	1\,804.3	0 %
A1	+ best observed reward	2\,246.1	0 %
A2	+ best observed reward, jump and swap	2\,461.8	0 %
C1	+ Metropolis accept/reject (lattice head)	3\,067.2	48 %
C1'	the same with a continuous head	2\,737.0	0 %
A0 evaluated deterministically (argmax)		1\,445.5	0 %

S6 Certificate gallery

Figure S3 shows one held out instance at fill ratio 0.95 solved by the same largest first constructor under the four certificate settings of the main text’s certificate comparison. Without a certificate the constructor occupies too much free space in the early steps, and the last facilities have no legal position; the scalar objective reports this only as a lower number.

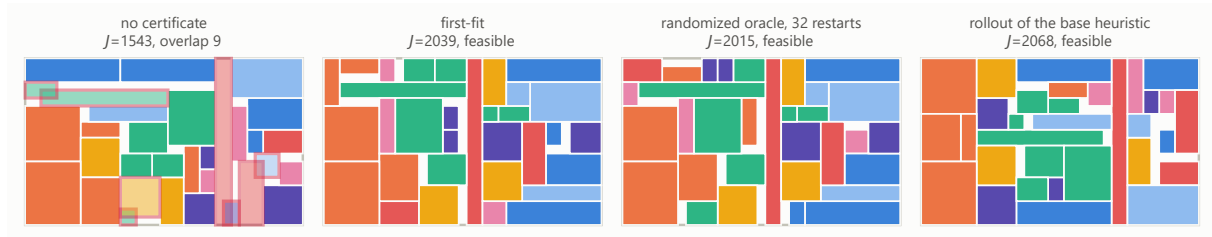


Figure S3: One instance, four certificates, same constructor. Facilities that end up overlapping are outlined in red.

S7 MCNC/GSRC conversion, in full

HPWL numbers depend on the conversion, so we state it completely. Block dimensions come from the GSRC `.blocks` files (`hardrectilinear` records, width and height from the corner list); netlists from the UCLA `.nets` files; terminal positions from the `.pl` files, matched to net pins by name. Pins are placed at block centers and terminals at their given coordinates, the standard simplification. Nets with fewer than two pins are dropped, contributing zero. Coordinates are divided by an integer for each instance and floored, which is necessary because the legality masks are dense over the plate: the factor is 6 for AMI33, 33 for AMI49 and 3 for the GSRC instances, chosen as the smallest divisor bringing the plate under 200 cells per side, and it is reported with every result. The outline is fixed by area, $\text{plate} = \text{total block area} / (1 - \text{dead space})$, at aspect ratio 1; blocks may be rotated by 90° . Under this conversion HPWL is comparable across the paper’s own methods but not against published floorplanning numbers, which use full resolution and a different pin model. Integer flooring changes block shapes and packing possibilities, so converted dead space is not treated as a published full-resolution result.

S8 Larger-instance constructor feasibility

Table S6 gives the full feasibility record of the constructors without training on 96 to 256 facilities summarized in the main text. These are constructor diagnostics; M1’s direct quality evaluation at 128 and 256 facilities is reported in S13.

Table S6: Constructors without training on larger instances, feasibility rate. Every instance has a verified feasible packing, so a failure indicates that the method did not find the existing packing. Plate geometry is drawn per cell, so methods are compared within each row. The rollout certificate is $O(n^2WH)$ per query and was not run above 128 facilities.

n	instances	no certificate	first fit	rollout
<i>fill ratio 0.85</i>				
96	15	1.00	1.00	1.00
128	15	1.00	1.00	1.00
192	15	1.00	1.00	not run
256	15	1.00	1.00	not run
<i>fill ratio 0.95</i>				
96	30	0.57	0.53	0.90
128	30	0.53	0.57	0.90
192	30	0.70	0.63	not run
256	30	0.83	0.83	not run

S9 One-second mechanism study

Every row below uses a common verified best-contact layout and retains it as a feasible incumbent. Stochastic seeds are averaged within an instance before cell means are computed. The 0.1 and 0.3 second values are read from the same anytime run that continues to 1.0 second. Times exclude the common measured initialization cost, which is charged equally to all methods in total-time reporting. Mean initialization and verification time ranges from 0.0052 to 0.0128

seconds over the four cells. All normalized improvements use the full feasible objective J_β , including the constant no-overlap bonus.

Table S7: One-second dense study at fill 0.95. Entries are mean percentage improvement over the common initial layout.

geometry	n	sec.	B1	B2	B2S	B3	M0	R0	M1
guillotine	32	0.1	1.72	0.55	0.82	2.45	3.50	1.64	6.14
		0.3	1.72	1.22	1.82	4.37	7.59	1.96	7.83
		1.0	1.72	2.40	3.51	9.63	9.04	2.43	9.51
guillotine	64	0.1	0.00	0.22	0.41	1.29	1.72	0.21	1.41
		0.3	4.20	0.68	0.87	2.21	3.64	1.33	5.21
		1.0	4.44	1.41	1.62	3.69	8.00	2.31	8.76
nonslicing	32	0.1	1.79	0.57	1.02	1.95	3.50	1.91	6.32
		0.3	1.79	1.43	2.03	4.13	7.63	2.27	7.83
		1.0	1.79	2.64	3.71	9.53	10.15	2.86	9.65
nonslicing	64	0.1	0.00	0.18	0.44	1.64	1.47	0.13	1.14
		0.3	3.46	0.74	0.96	2.31	2.56	1.46	4.68
		1.0	4.10	1.38	1.78	3.60	7.78	2.70	7.98

Table S8: Pooled M1 minus control contrasts on 196 initialized dense instances. Entries are percentage points with paired instance-bootstrap 95% intervals.

sec.	B1	B2	B2S	B3	M0	R0
0.1	2.84 [2.26,3.45]	3.32 [2.83,3.82]	3.03 [2.56,3.52]	1.87 [1.36,2.39]	1.18 [0.58,1.75]	2.75 [2.33,3.17]
0.3	3.55 [2.57,4.49]	5.35 [4.84,5.86]	4.95 [4.47,5.43]	3.12 [2.58,3.67]	1.05 [0.46,1.67]	4.61 [4.12,5.10]
1.0	5.93 [4.99,6.88]	7.02 [6.54,7.52]	6.33 [5.84,6.82]	2.41 [1.74,3.06]	0.25 [-0.42,0.92]	6.39 [5.89,6.92]

At 1.0 second, the raw M1 minus B1 difference is 224.5 objective units [172.2,276.5], its win/tie/loss count is 156/0/40, and 152 of 196 instances exceed the one-percent practical threshold. The corresponding counts against B2, B2S, M0 and R0 are 196/0/0, 195/0/1, 109/0/87 and 195/0/1. The M0 interval does not establish a difference.

Table S9: Boundary study at fill 0.90 and 1.0 second. Entries are mean percentage improvement over the common initial layout. The last column is the paired M1 minus B1 percentage-point difference with a 95% bootstrap interval.

geometry	n	init.	B1	B2	B2S	B3	M0	R0	M1	M1-B1
guillotine	32	20/20	13.03	7.89	7.60	8.84	16.17	7.97	14.60	1.57 [-0.27,3.65]
guillotine	64	20/20	14.06	5.53	4.94	4.70	15.48	6.25	13.44	-0.62 [-2.09,1.07]
nonslicing	32	19/20	11.86	6.92	7.28	9.43	14.90	8.03	13.84	1.97 [-0.07,4.02]
nonslicing	64	20/20	14.98	5.01	4.82	4.10	15.85	6.68	13.07	-1.91 [-3.54,-0.25]

Across all dense M1 runs, the repair branch produces 6,215 incumbent updates and 142,270 objective units of logged gain; unchanged reuse produces 1,380 updates and 12,455 units. These event counts are mechanism descriptions rather than inferential samples. B2 and B2S have mean destroy sizes 8.88 to 9.06 and 2.99, respectively.

All 8,820 dense budget layouts and 5,214 boundary budget layouts pass an independent result audit. Recomputed objective and trace selection errors are zero. No admitted dense

Table S10: Mechanism log at fill 0.95 and 1.0 second. Repair success is conditional on a nonempty displaced set with size at most four. Verified completions exclude the carried fallback. R0 contains reuse only, and M0 uses full regeneration for every nonfallback proposal.

geometry	n	mean $ D $	repair success	M1 reuse	M1 verified	R0 verified	M0 verified
guillotine	32	1.95	2.45%	14.0	80.8	26.4	59.6
guillotine	64	2.49	3.91%	6.4	61.1	23.8	40.9
nonslicing	32	2.05	2.19%	15.9	76.1	29.7	59.0
nonslicing	64	2.72	3.55%	6.9	55.0	26.2	35.7

Table S11: Mean measured work per one-second dense run, in milliseconds. Late is the number of candidates that completed after the cutoff and were discarded.

method	generation	intermediate verify	final verify	objective	return time	late
B1	0.0	0.0	0.1	0.1	1001.53	0
B2	0.0	0.0	2.9	2.6	1002.58	1
B2S	0.0	0.0	15.6	13.5	1002.67	14
B3	845.4	37.7	40.3	40.9	1001.44	28
M0	719.3	21.4	30.7	30.6	1001.26	23
R0	0.0	12.6	105.0	92.9	1001.66	81
M1	358.1	30.3	65.2	61.8	1001.11	47

event is later than 0.999998 seconds. Nonpreemptive work can return up to 9.8 ms after the cutoff, but its candidate is not admitted. The adversarial regression set rejects duplicate and missing facilities, negative and out-of-range coordinates, overlap, stale witnesses and incomplete timeout returns, while accepting legal edge contact.

S10 Additional formulation and certificate diagnostics

These experiments explain the formulation and certificate choices that motivate completion-aware construction. Table S12 shows why an overlap penalty receives little feasible exploration as size and density increase. The lattice column is higher than the continuous-center column wherever both are resolved, so action representation matters before learning.

Table S12: Fraction of uniformly sampled layouts that are overlap free. Each entry averages four generated instances at 4×10^4 samples; $< 6 \times 10^{-6}$ means no feasible sample was drawn.

n	fill 0.35		fill 0.50		fill 0.65	
	continuous	lattice	continuous	lattice	continuous	lattice
2	2.6×10^{-1}	3.7×10^{-1}	1.7×10^{-1}	2.4×10^{-1}	7.1×10^{-2}	1.4×10^{-1}
4	3.3×10^{-2}	7.6×10^{-2}	3.0×10^{-3}	1.5×10^{-2}	5.0×10^{-4}	3.9×10^{-3}
6	4.0×10^{-3}	1.7×10^{-2}	1.6×10^{-4}	1.1×10^{-3}	$< 6 \times 10^{-6}$	2.5×10^{-5}
8	5.9×10^{-4}	4.5×10^{-3}	1.9×10^{-5}	6.9×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
10	8.8×10^{-5}	1.1×10^{-3}	$< 6 \times 10^{-6}$	1.3×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
12	6.3×10^{-6}	2.2×10^{-4}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
16	$< 6 \times 10^{-6}$	1.3×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$

The single-facility translation baseline uses the reward $J(s_{t+1}) - J(s_t)$. With discount one it optimizes the terminal objective; with discount below one it favors high intermediate values and

penalizes recovered detours. At a common 688k-step training budget, adding a best-observed reward, jump and swap moves, and Metropolis acceptance raised the objective from 2 658.5 to 3 067.2 and feasibility from 0% to 48%. Lattice annealing reached 3 414.8, and a masked constructor without training reached 3 380.2. Table S5 gives the complete formulation ablation.

Initialization coverage depended strongly on the witness generator. On 100-instance dense cells $(n, f) = (32, 0.90), (32, 0.95), (64, 0.95)$, bottom-left fill in area order covered 0.78, 0.09 and 0.28; best-contact covered 0.98, 0.96 and 1.00. To control for initialization, every primary method receives the same verified best-contact layout. In the full-regeneration decoder diagnostics, the exact marginal selector matched or exceeded the two learned selectors in most $n = 64$ cells, supporting M1’s use of exact marginal ranking.

Finally, the learned completion predictor was trained on 25,258 rollout states with an instance-grouped split. Validation AUC was 0.902, but on disjoint fill 0.95 tests its best completion rate was 0.20, compared with 0.90 for verified first fit and 1.00 for the more expensive rollout certificate. The labels record the outcome of one heuristic rollout; membership in the set of completable states requires a feasible continuation. These results motivate using learned scores for candidate ordering together with independent feasibility verification.

On the two hand-authored scenarios, masked constructive search supplies most of the improvement over translation-based search, but learning adds only 0.9% over the best training-free procedure in the same action set. On 60 held-out moderate-density instances, beam search without training has the best paired objective and solves 57, compared with 60 for the policies. These results support the use of training-free selectors in the primary comparison; supporting tables and complete run-level tests are reported in the Online Supplement and archive, respectively.

S11 Secondary methods and evaluation protocol

These experiments supplement the primary partial-repair comparison.

Design The suite crosses four facility counts, $n \in \{8, 16, 32, 64\}$, with three fill ratios, $f \in \{0.40, 0.60, 0.75\}$, at 20 instances per cell: 240 instances. Five extensions follow the same construction from disjoint seeds. the density diagnostic adds $f \in \{0.80, 0.85, 0.90, 0.95\}$ at $n = 32$ (80) and $f \in \{0.85, 0.90, 0.95\}$ at $n = 64$ (60); the large-instance diagnostic adds $n \in \{96, 128, 192, 256\}$ at $f = 0.85$ and $f = 0.95$, with plates enlarged in proportion so facilities do not shrink to the minimum size (180). The certified sweep also covers the $n = 64, f = 0.80$ cell (20). Finally, the frozen decoders are run on a confirmatory family of 100 held out instances in each $n \in \{32, 64\}$ by $f \in \{0.90, 0.95\}$ cell (400), and a small plate family at $n = 8$ (60), sized so that constraint programming can prove optima, supports the gap study of the main-text small-instance calibration and moderate-density diagnostic. These diagnostic suites contain 1 040 distinct held out instances. The one-second mechanism study adds 200 dense instances and 80 boundary instances. The 60-second study adds 240 instances under distinct tags, including 128- and 256-facility cells (S13). Every source packing is verified. Achieved fill is recorded for each instance; integer dimensions make the target approximate. Plate aspect ratio, types, affinities and boundary preferences vary by instance. The randomized objective constants are drawn once per suite cell.

Splits Train, validation and test draw from disjoint generator seed ranges. Instance identity is determined by the family, split, facility count, nominal fill, batch size and index, together

with the stored seed and tag. The policy confirmation and the primary comparison families use distinct generator tags. These test families are mutually disjoint and are disjoint from training, validation and the other test cells.

S11.1 Diagnostic methods

The matched-time methods B0 to B3, B2S, M0, R0, M1, and ALNS are defined in the main-text method table. They are classical CPU implementations and share the initial layout, objective, time limit and verifier. The following methods are used in the diagnostic experiments.

- **Classical search:** shelf packing; lattice constrained simulated annealing at 8 192 and 45 000 proposals; tabu search over single facility moves; and a lattice genetic algorithm with uniform crossover and elitism. Annealing and tabu use periodic greedy polish and a common budget of 45 000 objective calls. Constraining annealing to the lattice is itself worth 4.4% on OFFICE and raises CLINIC feasibility from 20% to 100%; we report against this stronger version. Tabu’s tenure and start and the GA’s population and memetic setting are chosen on validation like every other hyperparameter: tabu keeps the constructive start (steepest single facility moves cannot repair a random dense start), and the memetic variant was rejected for exceeding the evaluation budget tenfold without improving on its compliant counterpart.
- **Masked constructive methods without training:** greedy with largest first order; greedy with joint choice of facility and position; regret k insertion for $k \in \{2, 4, 8\}$; beam search over insertion decisions at widths 1 to 64; and beam search with the order fixed to largest first.
- **Methods without training and with certificate guidance:** largest first and joint greedy with the certificate filter of the main-text witness construction and an argmax fallback. These are empirical heuristics, not instances of the main-text feasible-continuation theorem.
- **Learned:** a constructive policy trained with PPO (Schulman et al., 2017); its heuristic variant with certificate guidance; and its full-regeneration certified variant with a carried witness. This last method uses the invariant of the main-text feasible-continuation theorem, but not the partial repair of the main-text partial-repair algorithm. Architecture and hyperparameters are in S12.

Beam width and regret k are selected on the *validation* split for the objective comparison of the main-text small-instance calibration and moderate-density diagnostic, and the test split is evaluated once with them fixed. The density cells of the density diagnostic report the whole sweep instead, because the finding there is the pattern of collapse across widths. On the two reference scenarios, single instances with no validation counterpart, the best setting is taken on the scenario itself; that favors the baselines without training, and we flag it there rather than reporting a p value.

S11.2 Policy coverage

The encoder concatenates pair features of length n , so one checkpoint serves one facility count (the main-text limitations). No single policy spans the suite, and Table S13 states exactly which

Table S13: Datasets, and the policies evaluated on them. Training budgets and wall clock differ by policy family and are listed per checkpoint in Table S14 of S12; training cost is per policy, not amortized across rows.

dataset	n	inst.	policies	trained on
generated, main	8, 16	120	none	not used
	32	60	3 seeds, env order	train, $f \leq 0.75$
	64	60	see density rows	not used
generated, density	32	80	2 seeds, own order	train, $f=0.75$ to 0.90
	64	60	2 seeds, own order	train, $f=0.75$ to 0.90
generated, certified only	64	20	2 seeds, own order	frozen; $f=0.80$
generated, dense confirm	32, 64	400	2 seeds, own order	frozen checkpoints
same initial layout, dense	32, 64	200	none	not used
same initial layout, boundary	32, 64	80	none	not used
generated, large	96 to 256	180	none (fixed n)	not used
generated, exact	8	60	none	not used
reference scenarios	32	2	1 per scenario	that scenario
MCNC / GSRC	33 to 300	5	none (other obj.)	not used
QAPLIB	12 to 30	32	none (other obj.)	not used

instances a learned method was run on and which are comparisons of methods without training. Aggregate instance counts elsewhere in the paper refer to held out *test* instances; training and validation instances are drawn from disjoint seed ranges and are not included in the reported test aggregates.

The diagnostic experiments use the following protocol. Every method is attempted on every instance in the cells assigned to it. Objective comparisons are paired only on instances for which both methods return feasible layouts. We normalize by the best *feasible* objective achieved on that instance, because raw values differ by an order of magnitude between cells with 8 and 64 facilities. We report the mean paired gap, a 95% bootstrap confidence interval over instances, a Wilcoxon signed rank test on paired differences with Holm correction across the family, and the number of instances on which each method wins. For binary reliability we report exact binomial confidence intervals. In the prespecified dense confirmation, guided and unfiltered decoders are paired instance by instance using the exact two sided McNemar test, with Holm correction across all twelve selector and cell comparisons. Coverage and feasibility conditional on return are analyzed separately for the selective decoder.

S12 Policy and training

Facility tokens are embedded from features for each facility (size, placed flag, position, boundary coefficients, and the affinity and distance rows against all facilities) and mixed by self attention; a pointer head selects the facility. Position selection is a fully convolutional score over the plate from dilated residual convolutions on seven planes (occupancy, the legality mask, the affinity of the facility being placed toward whichever facility owns each cell, the boundary bonus it would earn, a mask for positions within bounds and two coordinate planes), modulated channelwise by the selected facility’s embedding. The encoder has no separate embedding for each facility index, which permits transfer across briefs at fixed n . The reward is the exact marginal of the main-text local marginal proposition, so the episode return is the objective of the completed layout up to a constant.

Table S14: Training runs for the learned methods, from the stored checkpoint histories and predictive filter metadata. Budgets end at an iteration boundary, so nominal 1.2M budgets record as 1,196,032 or 1,198,080.

policy or configuration	role	budget	min	deviations
Dense-32, seeds 1/2	density, $n = 32$	1 196 032	34/50	
Dense-64, seeds 1/2	density, $n = 64$	1 196 032	34/34	
Suite, seeds 1/2/3	main suite	1 196 032	45/44/42	random order
Transfer	transfer (S1)	1 198 080	17	64 env, 2 ep, mb 256, 48 ch
Office	OFFICE	1 998 848	34	2M steps
Clinic	CLINIC	899 072	7	0.9M steps, 64 env
A0/A0'/A1/A2/C1	improvement ablation	688 128	3.2 to 3.4	five separate runs
C1'	continuous head (S4)	688 128	2.5	continuous displacement
Completion predictor	predictive filter diagnostic	4 000 ep.	0.28	grouped validation

Unless a row of Table S14 says otherwise, every constructive policy shares one PPO configuration: $\gamma = 1$, GAE $\lambda = 0.95$, clip 0.2, Adam at 3×10^{-4} , 3 epochs per batch, minibatch 512, value coefficient 0.5, gradient norm clip 0.5, entropy coefficient 0.01 decayed linearly to zero at 80% of training with a floor of 5% of its initial value, token width 128, three attention blocks, 64 position head channels and 128 parallel environments. For every policy, the final checkpoint is used; no checkpoint selection based on validation is performed. The predictive filter is a separate supervised diagnostic: it retains the lowest validation loss state over 4 000 epochs and fixes its seven thresholds on that same grouped validation set before any test evaluation. The five improvement MDP checkpoints of the secondary formulation study (tabulated in the Online Supplement, S5) are trained separately at a matched budget of 688,128 steps (193 to 203s each). Wall clock for policy training is on one RTX 4080 SUPER; the predictive filter was trained on CPU in 17s.

The repository README maps these policy labels to the checkpoint files and training histories. Seed identifiers distinguish independently trained policies; the improvement configurations retain the identifiers used in the ablation.

S13 Comparison through 60 seconds

This study evaluates M0, M1, and an adaptive repair control on a common initialized set. It crosses $n = 32, 64, 128, 256$, guillotine and nonslicing geometry, and nominal fills 0.90 and 0.95, with 10 and 20 attempted instances per geometry and size cell at the respective fills. Instance tags are disjoint from S9, and these timing records are analyzed separately. The maximum plate side is 44, 44, 72, and 100 for the four sizes; sampled mean-area bounds remain 9 to 30. Integer dimensions make fill approximate. The generated source packing is verified but withheld from every search.

S13.1 Adaptive repair control and validation

The control is a layout-specific ALNS adaptation (Ropke & Pisinger, 2006). It combines three removal rules with two insertion rules. Random removal samples facilities uniformly without replacement. Spatial removal selects facilities with centers nearest a random anchor. Low-contribution removal ranks facilities by incident pair and boundary rewards and samples without replacement with weights proportional to the inverse square root of rank. The constant no-overlap bonus is omitted from these contribution scores. The removal count is uniform from two to the selected cap.

Beam insertion visits removed facilities in decreasing area order, shuffled with probability 0.25, and keeps a bounded frontier ranked by accumulated exact marginal gain. Regret insertion repeatedly inserts the facility with the largest gap between its two best legal positions, breaking ties by area. Both use the configured number of ranked candidate positions and append the facility’s original position when legal. Every complete candidate is checked by the same independent verifier and objective evaluator as M0 and M1.

The six operator-pair weights start at one. After an attempt, the selected weight is updated to $\max(0.2, 0.9w + 0.1r)$, where the reward is six for a new best layout, three for an accepted improving move of the current layout, one for an accepted nonimproving move, and zero otherwise. The best layout is retained separately from the current layout. A nonimproving move with change ΔJ is accepted with probability $\exp(\Delta J/T)$. The initial temperature is $0.0025 \max(|J(w_0) - \beta_{\max}|, 1)$ and is multiplied by 0.999 per attempt. These online updates are fixed parts of the search rule.

Validation uses 16 disjoint instances: two per geometry, fill, and size cell at $n = 64$ and 128, with two seeds and a 10-second cutoff. The configuration with the largest mean improvement after averaging seeds within instance is selected before testing. Table S15 reports all four candidates and the B2/B2S reference controls. The selected ALNS configuration has removal cap four, beam width four, and four ranked candidates. M0 and M1 retain $\kappa = 32$, one completion attempt, and M1’s repair cap of four.

S13.2 Coverage and quality

Table S16 includes all 240 attempted instances. Initialization succeeds on 236; the four failures are split equally between the two $n = 32$, fill 0.95 cells and are not replaced. Three methods and three seeds give 2,124 completed runs. Each run continues for 60 seconds and records cutoffs 0, 0.1, 0.3, 1, 3, 10, 30, and 60 seconds.

Tables S17 and S18 give geometry-specific quality at the primary cutoffs. The full set of nonzero cutoffs is shown in Figure S4. Improvement uses $100[J_m(t) - J(w_0)]/J(w_0)$, including

Table S15: Validation at 10 seconds on 16 instances with two seeds. Improvement is the mean percentage gain over the common initializer. The selected ALNS configuration has removal cap four and beam width four. Candidate counts exclude the extra original position when legal; B2 and B2S use a different insertion rule.

method	removal cap	beam width	candidates	improvement
ALNS	4	4	4	7.80
ALNS	8	4	4	7.63
ALNS	8	8	8	7.08
ALNS	16	4	4	7.15
B2	16	n/a	n/a	5.83
B2S	4	n/a	n/a	6.99

Table S16: Initialization coverage and achieved fill in the 60-second study. Failures remain in the attempted count. Fill ranges cover all attempted instances; mean initialization-attempt time is in seconds over all attempts.

geometry	n	fill	initialized	achieved fill	init. time
guillotine	32	0.90	10/10	0.89815 to 0.90000	0.0055
guillotine	32	0.95	18/20	0.94318 to 0.95000	0.0052
guillotine	64	0.90	10/10	0.89855 to 0.89959	0.0128
guillotine	64	0.95	20/20	0.94792 to 0.95000	0.0153
guillotine	128	0.90	10/10	0.89938 to 0.90000	0.0332
guillotine	128	0.95	20/20	0.94885 to 0.95000	0.0299
guillotine	256	0.90	10/10	0.89965 to 0.90000	0.0999
guillotine	256	0.95	20/20	0.94943 to 0.95000	0.0970
nonslicing	32	0.90	10/10	0.89583 to 0.90000	0.0044
nonslicing	32	0.95	18/20	0.94603 to 0.95000	0.0036
nonslicing	64	0.90	10/10	0.89836 to 0.90000	0.0099
nonslicing	64	0.95	20/20	0.94796 to 0.94994	0.0104
nonslicing	128	0.90	10/10	0.89890 to 0.90000	0.0274
nonslicing	128	0.95	20/20	0.94915 to 0.95000	0.0304
nonslicing	256	0.90	10/10	0.89942 to 0.90000	0.0905
nonslicing	256	0.95	20/20	0.94933 to 0.95000	0.0873

the constant feasible-layout bonus in the denominator. Three seeds are averaged within each instance. Pooled contrasts give equal weight to the two geometry families, with 10,000 paired bootstrap samples resampling instances within geometry. All 95% intervals are descriptive and pointwise, without multiplicity adjustment. Raw objective gains, every cutoff, and geometry-specific paired intervals are retained in the accompanying result tables.

Table S17: Geometry-specific mean percentage improvement at fill 0.90. Seeds are averaged within each instance before the cell mean. Time excludes initialization; all methods use the common initialized set.

geometry	n	seconds	M0	M1	ALNS
guillotine	32	10	25.08	21.96	20.00
guillotine	32	60	28.19	26.29	25.59
guillotine	64	10	17.49	14.80	12.95
guillotine	64	60	18.85	17.57	19.31
guillotine	128	10	16.20	14.31	8.52
guillotine	128	60	18.02	15.49	15.61
guillotine	256	10	8.66	13.44	3.28
guillotine	256	60	19.95	14.72	10.06
nonslicing	32	10	18.93	18.86	18.80
nonslicing	32	60	20.67	22.36	21.75
nonslicing	64	10	16.28	15.01	14.14
nonslicing	64	60	17.70	17.41	19.14
nonslicing	128	10	17.44	14.89	10.61
nonslicing	128	60	19.63	16.98	17.68
nonslicing	256	10	10.74	13.93	4.10
nonslicing	256	60	18.84	14.67	11.47

S13.3 Time to a target improvement

For each run and target $q \in \{1, 5, 10\}\%$, the first verified trace event reaching q defines the hitting time. A non-hit contributes the 60-second cutoff to the restricted mean. Table S19 reports both hit percentage and restricted mean, pooling geometries equally. A companion total-time column in the result archive adds each run’s measured initialization time. These summaries are descriptive, with equal seed counts per instance; no success-only timing average is used.

At $n = 256$, fill 0.95, M1 reaches a 5% improvement in 94.2% of runs with a restricted mean of 9.76 seconds, compared with 90.0% and 20.73 seconds for M0. At the 10% target, the order reverses: M0 reaches the target in 56.7% of runs with a restricted mean of 39.15 seconds, while M1 reaches it in 20.0% with 51.74 seconds. This separates early moderate improvement from attainment of a more demanding target.

S13.4 Timing and independent checks

Validation and test use eight single-threaded processes pinned to distinct physical cores of an AMD Ryzen 7 9700X, with Python 3.10.18 and NumPy 2.2.5 on Windows 10. All methods and seeds for one instance run on the same core in deterministic shuffled order. BLAS and OpenMP thread counts are one. Table S20 reports the full-run CPU/wall ratios and return overruns.

Table S18: Geometry-specific mean percentage improvement at fill 0.95. Seeds are averaged within each instance before the cell mean. Time excludes initialization; all methods use the common initialized set.

geometry	n	seconds	M0	M1	ALNS
guillotine	32	10	13.60	15.60	11.87
guillotine	32	60	17.72	19.38	16.25
guillotine	64	10	13.90	11.87	7.78
guillotine	64	60	16.08	15.90	13.99
guillotine	128	10	11.56	9.61	4.45
guillotine	128	60	15.50	12.07	9.14
guillotine	256	10	5.71	8.00	1.75
guillotine	256	60	11.80	9.12	4.34
nonslicing	32	10	14.74	15.44	12.26
nonslicing	32	60	17.44	18.24	15.43
nonslicing	64	10	11.89	11.91	7.67
nonslicing	64	60	13.65	14.87	13.41
nonslicing	128	10	9.52	9.52	4.48
nonslicing	128	60	13.15	11.88	8.65
nonslicing	256	10	3.82	7.32	1.76
nonslicing	256	60	8.96	8.03	4.56

Table S19: Target attainment within 60 seconds, pooling the two geometries with equal weight. Each entry is hit percentage / restricted mean search time in seconds. Non-hits contribute 60 seconds, so the time is not a success-only mean. Three seeds per instance contribute equally.

n	fill	method	target 1%	target 5%	target 10%
32	0.90	M0	100.0 / 0.02	100.0 / 0.05	100.0 / 0.12
32	0.90	M1	100.0 / 0.02	100.0 / 0.04	100.0 / 0.23
32	0.90	ALNS	100.0 / 0.06	100.0 / 0.37	100.0 / 1.81
64	0.90	M0	100.0 / 0.07	100.0 / 0.22	83.3 / 11.19
64	0.90	M1	100.0 / 0.05	100.0 / 0.51	95.0 / 6.15
64	0.90	ALNS	100.0 / 0.16	100.0 / 1.69	95.0 / 10.23
128	0.90	M0	100.0 / 0.21	100.0 / 1.17	93.3 / 7.35
128	0.90	M1	100.0 / 0.16	100.0 / 0.58	90.0 / 7.01
128	0.90	ALNS	100.0 / 0.46	100.0 / 3.79	95.0 / 16.17
256	0.90	M0	100.0 / 1.37	100.0 / 7.10	100.0 / 11.71
256	0.90	M1	100.0 / 0.64	100.0 / 2.58	100.0 / 5.30
256	0.90	ALNS	100.0 / 3.01	96.7 / 20.55	53.3 / 49.75
32	0.95	M0	100.0 / 0.06	100.0 / 0.65	88.0 / 12.76
32	0.95	M1	100.0 / 0.04	100.0 / 0.71	93.5 / 7.92
32	0.95	ALNS	100.0 / 0.13	99.1 / 2.80	83.3 / 17.19
64	0.95	M0	100.0 / 0.25	95.8 / 4.14	77.5 / 18.11
64	0.95	M1	100.0 / 0.11	100.0 / 0.37	90.8 / 14.55
64	0.95	ALNS	100.0 / 0.34	100.0 / 6.92	71.7 / 31.12
128	0.95	M0	100.0 / 1.87	94.2 / 7.48	72.5 / 23.71
128	0.95	M1	100.0 / 0.55	95.8 / 5.73	65.8 / 28.08
128	0.95	ALNS	100.0 / 1.21	87.5 / 22.48	31.7 / 51.49
256	0.95	M0	100.0 / 4.51	90.0 / 20.73	56.7 / 39.15
256	0.95	M1	100.0 / 1.96	94.2 / 9.76	20.0 / 51.74
256	0.95	ALNS	100.0 / 6.25	38.3 / 54.54	0.0 / 60.00

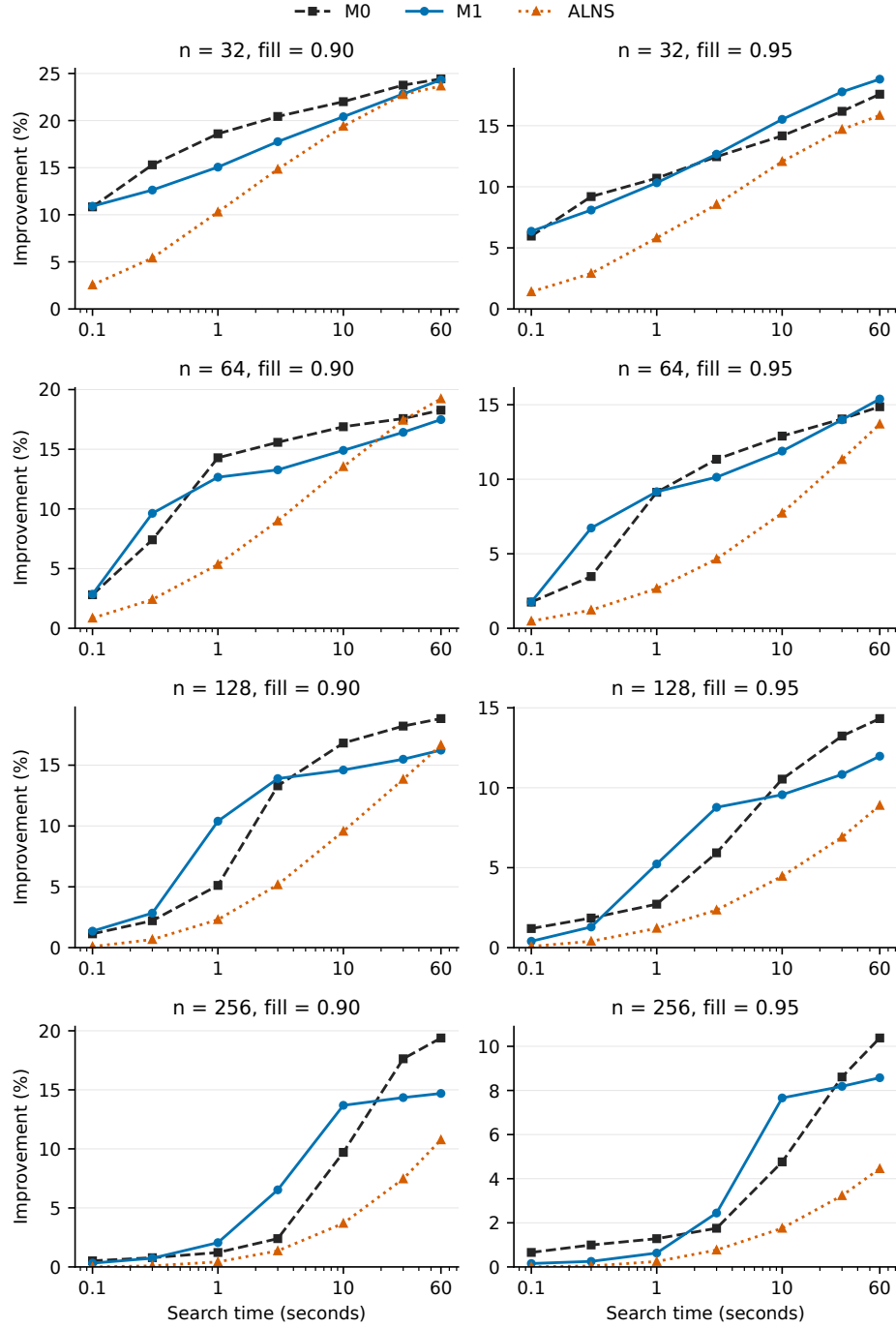


Figure S4: All size and fill cells of the 60-second comparison. Curves average seeds within instance and then give the two geometries equal weight. Connecting lines guide the eye between recorded cutoffs; they are not independently restarted searches. The common initialization time is excluded.

No low-ratio run is removed. These are throughput measurements under the stated allocation, rather than single-worker latency measurements.

Table S20: Timing over 708 runs per method. Ratios compare process CPU time with wall time. Return overruns are milliseconds beyond the 60-second deadline. Late completed candidates are discarded, not admitted to any quality comparison. Final verification failures are zero for all methods.

method	mean ratio	min. ratio	median ratio	mean overrun	max. overrun	late
M0	0.9798	0.6898	0.9970	5.00	61.06	42
M1	0.9794	0.6953	0.9971	3.08	43.61	74
ALNS	0.9805	0.6515	0.9973	8.44	66.43	14

The independent audit regenerates all attempted instances, checks their source packings and common initializations, verifies all 17,228 saved layouts, and recomputes every saved objective. It also checks the expected run identities, trace monotonicity, and the eligibility of every saved budget row. All checks pass, with zero objective recomputation error. The 130 completed events finishing after the final deadline are discarded. Protocol and result-manifest hashes link the reported tables to these audited records.

References

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